

SIMATS ENGINEERING
CO2_ASSESSMENT TOOL3_Application Based Questions

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

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COURSE OUTCOME COVERED

CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct

I A. Lakshmi(192373066)certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:Lakshmi

Assessment Tasks

Application Based Questions

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.
2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.

3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.
4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.
5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

Application Based Questions Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.

Cloud Auto-Scaling & Load Balancing in Seasonal E-Commerce

Imagine a company like **Flipkart or Amazon during Diwali sales**. Traffic can spike unpredictably—millions of users hitting the site at once.

- **Auto-scaling:** Cloud platforms (AWS EC2 Auto Scaling, Azure VM Scale Sets) automatically add more servers when demand rises. When traffic drops, they scale back down, saving costs. This elasticity ensures the site never crashes under load.
- **Load balancing:** Services like AWS Elastic Load Balancer or Azure Load Balancer distribute incoming requests across servers. If one server fails, traffic is rerouted instantly, preventing downtime.
- **Reduced downtime & improved UX:** Customers don't face "site unavailable" errors, checkout remains smooth, and pages load quickly. This directly translates to higher sales and customer trust.
- **Monitoring tools:** CloudWatch, Azure Monitor, or Datadog continuously track CPU, memory, and response times. They trigger alerts or scaling events before users notice issues.

Real-world scenario: During Black Friday, Amazon uses auto-scaling and load balancing to handle billions of requests. Without it, customers would face slow pages or failed checkouts, costing millions in lost revenue.

2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.

Cloud Storage for Secure Backups

Organizations generate massive amounts of data daily. Storing backups in the cloud ensures resilience and security.

- **Durability & redundancy:** Cloud storage (AWS S3, Google Cloud Storage) replicates data across multiple servers and regions. Even if one data center fails, copies remain intact. AWS S3 famously offers "11 nines" durability (99.999999999%).

- **Security:** Encryption protects data at rest and in transit. Access control (IAM policies, role-based permissions) ensures only authorized users can retrieve backups.
- **Cost advantage:** Traditional storage requires buying hardware, maintaining data centers, and handling upgrades. Cloud storage is pay-as-you-go, meaning you only pay for what you use.
- **Disaster recovery use case:** Suppose a company's primary data center is hit by a ransomware attack. Cloud backups allow them to restore clean data quickly, minimizing downtime and financial loss.

3.A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application

Global Deployment with CDNs

For a mobile app company launching globally, performance is critical. Let's take **Netflix or PUBG Mobile** as an example.

- **CDNs:** Content Delivery Networks (like Cloudflare, AWS CloudFront, Akamai) cache content on edge servers worldwide. Instead of fetching data from a distant origin server, users get it from a nearby location.
- **Reduced latency:** A gamer in Tokyo connects to a Tokyo edge server, not one in California. This reduces lag and improves responsiveness.
- **Edge servers:** They handle requests locally, reducing round-trip times and improving user experience.
- **Cloud provider support:** AWS, Azure, and Google Cloud provide global infrastructure, ensuring apps are available across continents with minimal downtime.

Real-world scenario: In real-time gaming, milliseconds matter. PUBG Mobile uses CDNs to ensure smooth gameplay across regions, preventing lag spikes that could ruin the experience.

4.A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.

Compliance & Security in Financial Institutions

Banks and financial institutions face strict regulations. Cloud providers design services to meet these compliance needs.

- **Regulatory compliance:** Providers adhere to standards like PCI DSS (for payments), GDPR (for data privacy), and HIPAA (for healthcare). They undergo regular audits to prove compliance.
- **Logging & monitoring:** Every transaction is logged. Tools like AWS CloudTrail or Azure Monitor provide audit trails, helping detect fraud or unauthorized access.
- **Encryption:** Sensitive financial data is encrypted both at rest and in transit. Even if intercepted, it's unreadable without keys.
- **Data sovereignty:** Institutions can choose specific regions to store data, ensuring compliance with local laws (e.g., keeping EU customer data within EU boundaries).
- **Example:** A bank processing credit card transactions ensures logs are auditable, transactions are encrypted, and customer data remains in-country to meet sovereignty laws.

5.A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

Containerization vs. Virtual Machines

Containers revolutionize application deployment compared to traditional VMs.

- **Difference from VMs:** VMs run a full OS for each instance, consuming heavy resources. Containers share the host OS but isolate applications, making them lightweight and faster to start.
- **Performance & resource usage:** Containers use fewer resources, allowing more applications to run on the same hardware.
- **Orchestration (Kubernetes):** Kubernetes automates deployment, scaling, and healing. If a container crashes, Kubernetes spins up a replacement instantly. It also balances workloads across clusters.
- **Portability:** Containers run consistently across environments—developer laptops, on-prem servers, or any cloud provider. This eliminates the “works on my machine” problem.
- **Example:** A microservices-based e-commerce app where the cart, payment, and recommendation engines each run in separate containers. Kubernetes ensures each service scales independently based on demand.